

Using SysV Initialization Process

SysV

- stands for System V
- older system to manage boot-up services
- originally from Unix System V operating system

init

- initial process (PID1)
- the `init` program determines which services to start based on **runlevels**
- runlevels are numbered **0** to **6**

Runlevel Functions

Numbered 0 to 6

runlevel

0	transitional level used to shut down system
1, s, S	single user mode like safe mode in windows
2	Debian distros → multi-user mode with GUI
3	Redhat → multi-user mode with CLI
4	undefined
5	Redhat → multi-user with GUI
6	reboot system

Runlevel Services

Found in `/etc/inittab`

syntax: `id:runlevels:action:process`

id	identification code
runlevel	application runlevels for this entry e.g. 345 means runlevels 3,4,5
action	action to be taken how to treat the process <i>wait</i> means start the process and wait for it to terminate
process	process to run

example

```
l0:0:wait:/etc/init.d/rc 0
l1:1:wait:/etc/init.d/rc 1
l2:2:wait:/etc/init.d/rc 2
l3:3:wait:/etc/init.d/rc 3
l4:4:wait:/etc/init.d/rc 4
l5:5:wait:/etc/init.d/rc 5
l6:6:wait:/etc/init.d/rc 6
```

Startup Scripts

1. the **rc** script runs scripts for each runlevel
2. symbolic links to scripts are placed in runlevel-specific directories
3. **rc** passes a **start** parameter to all the scripts with names that begin with **S**, and a stop parameter to all the scripts with names that begin with **K**

Managing Runlevel Services

chkconfig

manage SysV init service configuration

chkconfig options

options

<code>--list</code>	list all services as on/off and runlevels
<code>--list [service-name]</code> <i>chkconfig --list http</i>	show services as on/off and runlevel
<code>[service-name] on</code> <i>chkconfig http on</i>	enable service to start with default runlevel
<code>--level level [service-name] on</code> <i>chkconfig --level 23 nfs-common on</i>	enable with runlevels
<code>--add [service-name]</code> <i>chkconfig --add http</i>	add service

Checking Runlevel

Checking Default Runlevel

Check the `/etc/inittab` file

example

```
# grep :initdefault: /etc/inittab
id:2:initdefault:
```

the default runlevel has id:2

Changing Default Runlevel

Edit the **initdefault** line in `/etc/inittab`

if the system doesn't have that file, create one with only an **initdefault** line

Checking Current Runlevel

`runlevel`

example

```
# runlevel
N 2
```

N → THE system hasn't switched runlevels since booting

Changing Runlevels on a Running System

e.g. changing from GUI → CLI

you can use:

1. `init`
2. `shutdown`
3. `halt`
4. `reboot`

5. poweroff**init**`init runlevel`

example

`init 1` → change the runlevel to 1**telinit****available in SysV, Upstart, systemd**`telinit runlevel [options]`same as **init** but has an option to reload `/etc/inittab`

<code>q</code>	reload <code>/etc/inittab</code> and any changes
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example

`telinit q`**shutdown**`shutdown [options] [time] [message]`

sends a message to all users who are logged in

lets you specify when to change change runlevel, so users have to to save and exit

option	<code>now</code>	shutdown now
option	<code>-r</code>	reboot
option	<code>-P</code>	powers off
option	<code>-h</code>	halt, shutdown without powering off off
option	<code>-c</code>	cancel a pending shutdown
time	<code>hh:mm</code>	24 hour clock format

example

`shutdown -h +15 "system going down for maintenance"`**halt, reboot, poweroff****available in SysV, Upstart, systemd**`halt` shutdown without powering off`reboot``shutdown`